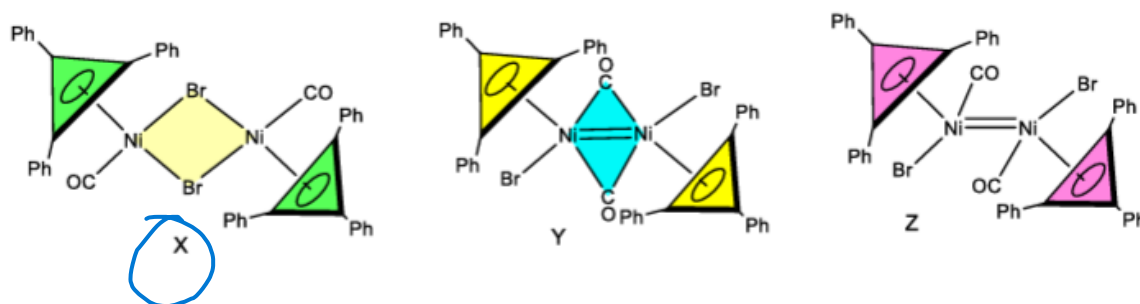


ACML 101 Inorganic Part
Home assignment -2
Organometallic Chemistry

Name: Entry No. Group.....

Please answer all the questions in the provided space and submit them by **December 20, 2024**.

(1) For the dimeric complex $[(\text{c-Ph}_3\text{C}_3)\text{NiBr}(\text{CO})]_2$ which one among the given three possibilities is the most likely structure? (The IR spectra of this compound gave a peak at 2039 cm^{-1}). Hints: It is generally observed that especially with first row transition metals, bridging ligands are favoured over metal-metal multiple bond formation for gaining 18 electrons.

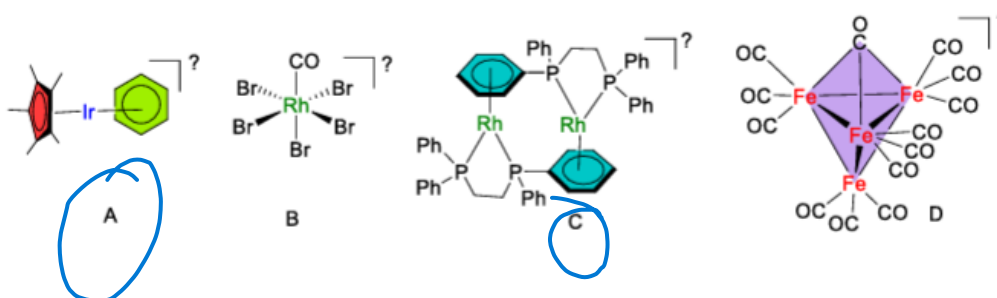


The correct option is X

Solution

Considering compliance with 18 e rule and nickel oxidation state (+2), all three structures seem possible. However, in reality, the compound has been found to adopt structure X having bridging bromines. It is generally observed that especially with first row transition metals, bridging ligands are favoured over metal-metal multiple bond formation for gaining 18 electrons. It is also observed that if anionic ligands and neutral ligands are present, most often the preference for bridging is more for the anionic ligand than the neutral ligand to gain 18 electrons as strain around the metal centre is also reduced in the process. The IR spectra of this compound gave a peak at 2039 cm^{-1} .

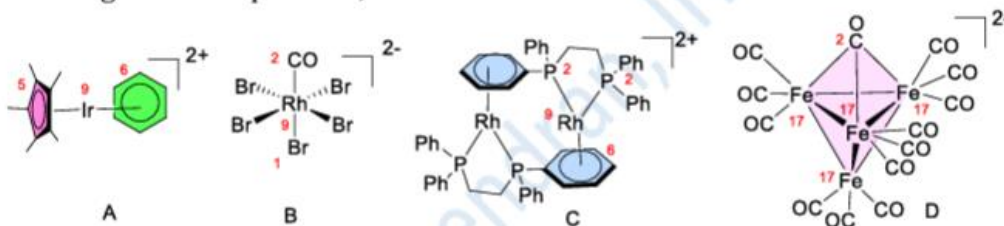
(2) The given cationic / anionic complexes obey the 18 electron rule. The complexes having 2^+ overall charge are... (Show the electron counts)



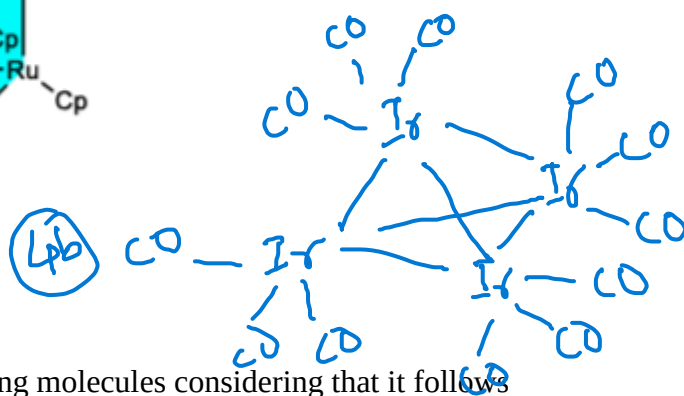
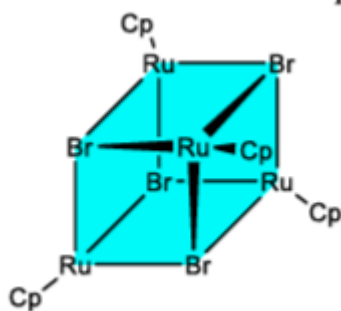
The correct option is A and C

Solution

The compounds having 2^+ overall charge are A and C (Electron count as per neutral atom counting method is provided).



(3) $[\text{Cp}^*\text{RuCl}]_4$ adopts a symmetrical structure having no metal-metal bonds where all the Ru centres have 18 electrons. Predict its structure ($\text{Cp}^* = \text{C}_5\text{Me}_5$)

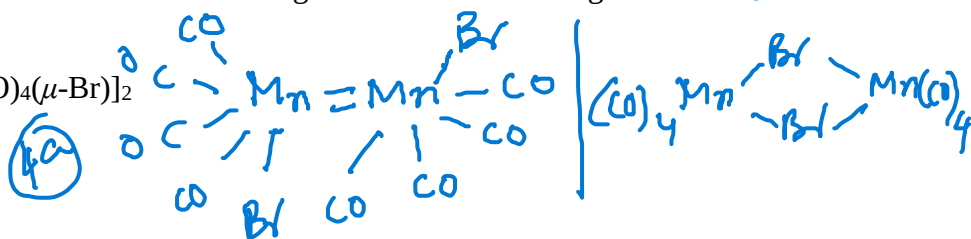


(4) Draw the most appropriate structure of the following molecules considering that it follows 18 electron rule

(a) $[\text{Mn}(\text{CO})_4\text{Br}]_2$ and $[\text{Mn}_2(\text{CO})_4(\mu\text{-Br})_2]$

(b) $\text{Ir}_4(\text{CO})_{12}$

(c) $[\text{HRu}\{\text{P}(\text{C}_6\text{H}_5)_3\}_3]^+$

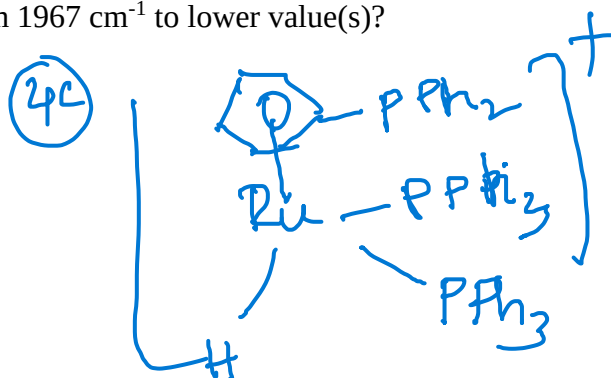


(5) For Vaska's complex, $\text{IrCl}(\text{CO})(\text{PPh}_3)_2$ which among the given activities will decrease the ν_{CO} stretching frequency of the coordinated CO ligand from 1967 cm^{-1} to lower value(s)? Justify your answer

(i) Replacing both PPh_3 by PMe_3

(ii) Oxidative addition by MeI

(iii) Replacing Cl^- by CH_3CN (cationic complex)



The correct option is (i)

where electron density on the metal gets increased by the activity. The actual ν_{CO} stretching frequencies are given below.

(i) Replacing both PPh_3 by PMe_3 : 1938 cm^{-1}

(ii) Oxidative addition by MeI : 2047 cm^{-1}

iii) Replacing Cl^- by CH_3CN (Cationic complex): 1975 cm^{-1}